major chemicals frequently present in WWTP effluents, which also show high toxicities. Kadokami et al. (1995) reported a technique for the simultaneous determination of 266 chemicals in water using GC/MS, an advanced chemical measurement, with more confined categories, should be suggested based on the frequency of detection and the degree of toxicity. Consequently, the biological method (i.e., WET testing), which measures the potential and real toxicity of mixed chemicals, should be used for monitoring the toxicity of WWTP effluents containing various hazardous chemicals.

References

Aguayo S, Muñoz MJ, de la Torre A, Roset J, de la Peña E, Carrballo M (2004) Identification of organic compounds and ecotoxicological assessment of sewage treatment plants (STP) effluents. Sci Total Environ 328:69–81

Barata C, Solayan A, Porte A (2004) Role of B-esterases in assessing toxicity of organophosphorous (chlorpyrifos, malathion) and carbamate (carbofuran) pesticides to *Daphnia magna*. Aquat Toxicol 66:125–139

Hsieh CY, Tsai MH, Ryan DK, Pancorbo OC (2004) Toxicity of the 13 priority pollutant metals to *Vibrio fisheri* in the Microtox® chronic toxicity test. Sci Total Environ 320:37–50

Kadokami K, Sato K, Hanada Y, Shinohara R, Koga M, Shiraishi H (1995) Simultaneous determination of 266 chemicals in water at ppt levels by GC-Ion trap MS. Anal Chem 11:375–384

Ma H, Kim SD, Allen HE, Cha DK (2002) Effect of copper binding by suspended particulate matter on toxicity. Environ Toxicol Chem 21:710–714

Paquin RR, Santore RC, Wu KB, Kavvadas CD, Di Toro DM (2000) The biotic ligand model: a model of the acute toxicity of metals to aquatic life. Environ Sci Policy 3:175–182

Persoone G, Goyvaerts M, Janssen C, De Coen W, Vangheluwe M (1993) Cost-effective acute hazard monitoring of polluted waters and waste dumps with the aid of Toxkits. ACE 89/BE 2/D3. Final Report, Commission of European Communities

Plette ACC, Nederlof MM, Temminghoff EJM, Van Riemsdijk WH (1999) Bioavailability of heavy metals in terrestrial and aquatic systems: a quantitative approach. Environ Toxicol Chem 18:1882–1890

Ra JS, Lee BC, Chang NI, Kim SD (2006) Estimating the combined toxicity by two-step prediction model on the complicated chemical mixtures from wastewater treatment plant effluents. Environ Toxicol Chem 25:2107–2113

Sarakinos HC, Bermingham N, White PA, Rasmussen JB (2000) Correspondence between whole effluent toxicity and the presence of priority substances in complex industrial effluents. Environ Toxicol Chem 19:63–71

State of Michigan (2002) Whole effluent toxicity regulation in Michigan. Department of Environmental Quality. Lansing, MI

US Environmental Protection Agency (1991) Technical support document for water quality-based toxics control. EPA/505/2–90-001. Office of Water, Washington

US Environmental Protection Agency (1993) Methods for measuring the acute toxicity of effluents and receiving waters to freshwater and marine organisms (4th edn). EPA/600/4-90/027F. Office of Research and Development, Cincinnati